

BakeEase: Redefining Baking Experiences with AI-Powered Multimodal Systems

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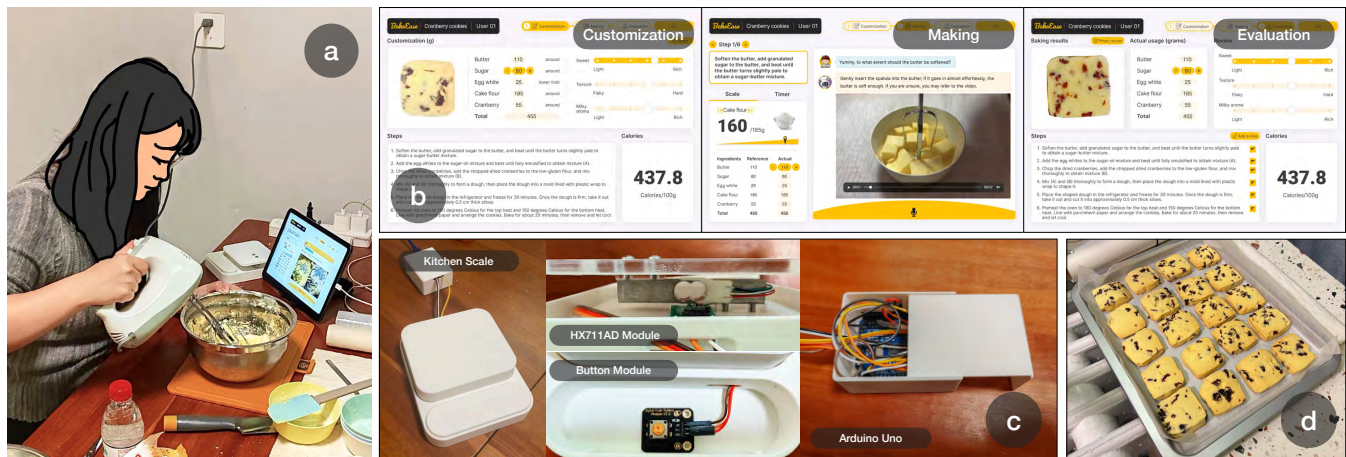


Figure 1: BakeEase helps baking enthusiasts complete baking tasks by guiding them through customized recipes with an AI assistant. (a) A user is baking cranberry cookies with BakeEase, interacting with its (b) multi-modality user interface and (c) kitchen scale. (d) With BakeEase's assistance, the user successfully achieves the desired baking results.

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Abstract

For baking enthusiasts, hands-on baking is essential for experiencing autonomy, creativity, and accomplishment. However, due to a lack of relevant knowledge or skills, they often follow fixed recipes, making it difficult to gain an in-depth understanding of baking principles to control the process entirely. To address this, we designed a hardware-software system, BakeEase, that enables baking enthusiasts to adjust recipes according to their preferences beforehand, integrates a chatbot and a large language model (LLM)

to make real-time ingredient adjustments during the process, and interacts multi-modally with the recipe to ask targeted questions about challenges encountered during baking. Additionally, users may record and evaluate the baking process and results afterwards. To explore how to design this system, we conducted observations and interviews with baking enthusiasts, followed by user testing the system prototype to gather feedback. The study results demonstrate that our system assists users in completing baking tasks while enhancing their overall baking experiences.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design.**

Keywords

cooking systems, chatbot-LLM integration, smart kitchen, smart home

ACM Reference Format:

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1 INTRODUCTION

Home baking is becoming increasingly popular worldwide, and the demand for homemade baked goods is growing [11]. Making delicious dishes at home brings comfort and joy and emphasizes healthy eating habits [27]. Especially during the COVID-19 pandemic, home baking helped people better cope with stress, find comfort, and connect with others [29, 38]. For baking enthusiasts, homemade food is a leisure activity, obtaining rich positive experiences (such as enjoyment, satisfaction, pride, creativity, a sense of control, and self-expression) during the process is essential [19, 39].

However, baking is a skill that emphasizes craftsmanship and has a learning curve. Mastering baking requires individuals to invest significant time to gradually accumulate specialized knowledge and practical experience to correctly use ingredients, tools, and equipment and master complex techniques. When encountering problems, enthusiasts need to sort through large amounts of information on social media to find relevant and practical content [26], which not only increases the difficulty of problem-solving but can also lead to information overload or incorrect learning. Additionally, the lack of professional guidance and feedback may confuse or frustrate novice bakers, hindering their skill development.

In this work, we developed the BakeEase system, which combines the advantages of a chatbot and a large language model (LLM) to design an innovative hardware-software system that meets users' diverse needs and provides comprehensive, flexible cooking support. Before cooking, users can adjust the recipe based on their taste preferences and health needs within certain thresholds using BakeEase. During baking, users can interact with the AI (Artificial Intelligence) assistant, Yummy, with natural language to control the user interface and a kitchen scale, record the actual ingredient amounts, adjust the ingredient states based on the recipe while

weighing, and seek targeted assistance from the Yummy when facing issues. After baking, users can review, evaluate their results, and add notes for sharing and future improvements.

We conducted a user study observing their baking processes and conducting semi-structured interviews to identify their challenges during baking. Based on these insights, we designed and implemented the system. Subsequently, we recruited five baking enthusiasts to use BakeEase and provide feedback. User experiments showed that BakeEase helps users complete baking tasks based on personalized recipes and improves user interaction through a voice-controlled interface, especially when their hands are occupied. Some participants showed differences in interaction preferences and offered suggestions for functional optimizations.

This paper's contributions are described below. First, we systematically identify and explain baking enthusiasts' behavioural characteristics and needs. Second, we propose a software system solution that combines hardware, chatbot, and LLM. This system aims to balance baking's autonomy with recipes' fixed limitations. Finally, our work integrates valuable user insights, discusses future system optimization directions based on their feedback, and advances intelligent cooking practices by addressing current challenges and presenting novel technical solutions supported by real-world feedback.

2 RELATED WORK

2.1 From Cookbooks to Cooking Systems

Traditional cooking recipes are usually step-by-step, using static content like images and text for guidance [7]. The shift from websites and blogs to apps (e.g., Kitchen Stories) and social platforms (e.g., TikTok) has diversified recipe presentation. However, these digital formats limit content and fail to address all user challenges. Enthusiasts often sort through social media for relevant content, reducing information efficiency [26]. In academic research, researchers have conducted extensive studies on cooking systems from various perspectives, such as optimizing the overall workflow of the kitchen [15, 16], recording and demonstrating cooking experiences [34], and sharing recipes among different smart kitchens [32]. Interactive cooking systems have also effectively assisted elderly individuals and the vulnerable in cooking activities [4, 6, 22, 23, 35]. In cooking systems, recipes play a crucial role. Many smart kitchen design projects have attempted to transmit or display recipe-related instructions or cooking information using digital interfaces, such as Counter Active [20] and Cooking Navi [13]. Previous research has shown that cooking systems that integrate recipes with multimodal interaction technologies for providing instructions can effectively assist users in completing predefined cooking tasks naturally [31]. Multimodal cooking systems like Cookie integrate voice assistants with software and hardware systems while enhancing adaptive solutions. These solutions can increase users' trust and acceptance of assistants [36]. Hardware-integrated systems like the Cooking Assistant System help provide contextual information, reducing user input and minimizing operational errors [30]. However, most of these studies focus on designing systems based on sequential workflows and fixed recipes, which do not sufficiently support users in adjusting ingredient quantities beforehand or during the cooking process. Thus, they overlook the dynamic nature of cooking.

2.2 Chatbot and LLM-Powered Cooking Assistants

In the HCI community, AI assistants for cooking have several precedents. For example, Chef Dalle uses a chatbot powered by LLMs to generate personalized recipes [14]. AREasyCooking recommends recipes based on dietary restrictions [17], while the Smart Cook system acquires ingredient information through multimodal interactions [25]; furthermore, Smart Recipes generates applicable recipes based on shopping lists [33]. These systems consider users' dietary restrictions and recommend recipes according to their personalized preferences. However, the chatbot can only recommend recipes from a pre-stored database and does not support users in adjusting recipes according to taste preferences, help users understand the knowledge and techniques embedded in recipes, or further assist in transforming external information into embodied knowledge. VUI (Voice User Interface) applications driven by Voice Personal Assistants (VPA) and supported by chatbots and LLMs hold significant potential for kitchen environments [24]. Studies have already demonstrated that dialogue-based interactions based on known recipes exhibit high accuracy in open-ended cooking processes [8] and conversational agents capable of engaging in dialogue significantly enhance the cooking experience [28]. Additionally, research has shown that LLM-supported systems can facilitate knowledge transfer during cooking, establishing a more engaging and efficient kitchen interaction experience [9], while transforming external information into embodied knowledge is crucial for cooking creativity [3].

In this work, we create a pre-stored database tailored to user needs, combining chatbots with LLMs, which draws knowledge from broader domains to provide complementary and targeted Q&A solutions. A connected kitchen scale allows the AI assistant to access contextual information. This study focuses on baking as a case study to develop and test BakeEase, exploring the potential of intelligent cooking.

3 DESIGN AND IMPLEMENTATION

3.1 Design Requirements Analysis

The needs study was conducted with seven baking enthusiasts (7 females, ages 20 to 36) recruited from the local community. The participants self-reported their baking skill proficiency with an average score of 2.57 (1 = very unfamiliar, 3 = moderate proficiency, 5 = high proficiency) and baking knowledge with an average score of 2.57 (1 = novice, 3 = some knowledge, 5 = extensive knowledge). Participants performed a cranberry cookie baking task on a living lab platform [10] using paper recipes provided by researchers. They could ask questions during the process to resolve difficulties. Semi-structured interviews followed to explore reported challenges and behaviour variations. Qualitative video analysis [2] and thematic analysis [5] were used to analyze the data. The cranberry cookie recipe was chosen due to its simplicity and the essential baking skills it encompasses.

Adjustment of ingredient proportions. Nearly all participants (N=6) made subjective adjustments to the amounts of butter and sugar in the recipe before or during baking. *"Seeing so much butter... make these few cookies, you feel like, oh, it seems quite an*

anxiety..." (P1). The participant is uncertain when adjusting the recipe. *"When I was making something before, and then I had to reduce the sugar, but the girl who was helping me said sugar couldn't reduce too much... I'm afraid it might affect emulsify..."* (P6).

Help and support. During the experiment, participants sought guidance from researchers on knowledge-based questions regarding the state of ingredients, including the degree of butter softening, whipping, and emulsification, etc. They also seek help with techniques, including how to mix the batter more quickly, slice thickness and plating methods. *"What was not very clear was, one, to what extent to whip it, um, another was emulsification, um, and then the way it mixes gives you much trouble"* (P2). Participants also inquire about the tool's purpose, usage methods.

Information acquisition. P4, P6, and P7 chose to watch videos while performing the tasks, and P6 used an app to seek up uncertain points in the recipe. *"I wanted to see what emulsification looks like, so I went to look it up on REDnote, but there weren't any..."* (P6). P2 mentioned the difficulties of acquiring essential information support. *"I don't really like searching for such specialized questions online... I feel like it's very hard to find this kind of information..."* (P2).

Assistance and feedback. During the experiment, participants requested assistance from researchers with operations such as pre-heating the oven, recording ingredient quantities, making notes and setting timers, while their hands were occupied. After completing key stages, the participant will seek positive confirmation from the researcher. *"...In this case, I'll start by filling in the blanks myself. Umm, if I think my argument makes sense, I'll ask the examiner... If I cannot understand, I will consult the examiner"* (P2).

By observing user behaviour and interviews, we found that baking enthusiasts need to overview the recipe before starting. They would like to make subjective adjustments to ingredient proportions, particularly in the amounts of sugar and butter. However, they are uncertain how these adjustments will affect the final baking outcome. Users who adjust the ratios of sugar and butter are primarily concerned with calorie intake which may make them overweight. During baking, enthusiasts seek information on operational techniques and baking knowledge from assistants or electronic devices. Most baking enthusiasts (N=6) weigh ingredients while adjusting the mixture's consistency rather than measuring everything once before starting. As a result, the kitchen scale emerges as a core interactive device. After baking, all participants tasted their final products and evaluated the modifications they made to the recipe. Their feedback mainly focused on how sugar and butter content changes affected sweetness, crispness, and buttery aroma.

Based on these insights, we designed the BakeEase software interface around three key stages: customizing the recipe before baking, providing assistance during the baking process, and evaluating the final product. We integrated the most frequently used kitchen scale with the system to accommodate task flow preferences: measuring first before baking or weighing ingredients while baking.

3.2 Prototype Design

BakeEase is a cooking assistance application that uses an AI-based software system with a kitchen scale to enable hands-free interaction and deliver real-time baking instruction via a Graphical User Interface (GUI) and Voice User Interface (VUI). The system consists of three primary components: a tablet that serves as a display and interaction device for the application, a computer that communicates with Azure and ChatGPT servers while running the Rasa framework, and an electric kitchen scale that provides weight data to the computer (see Figure 2). We categorized the common questions raised by participants based on Section 3.1, covering topics such as ingredient states and their causes, fundamental operational techniques, and tool usage methods. We constructed a dataset with the recipe and 36 Q&A in text, supplemented with images and video clips where necessary. Additionally, we incorporated 20 frequently used commands for controlling the GUI and kitchen scale. A pre-stored dataset was established to provide corresponding responses. ChatGPT answers questions about making cranberry cookies outside the dataset.

3.2.1 Hardware Design. The kitchen scale has a maximum capacity of 3kg and includes mass sensing and tare (zeroing) functions. For mass perception, we use a full-bridge 3kg pressure sensor and a HX711AD module positioned beneath the scale platform, as shown in Figure 1-b. Users may also use a button module to perform taring operations. Meanwhile, we have integrated a contactless engagement for taring via voice control with the AI assistant.

3.2.2 Interface Design. The BakeEase's user interface, as shown in Figure 3, consists of five main components: (a) a recipe panel that allows users to adjust ingredient ratios and preview outcomes using numerical inputs or sliders; (b) a multimodal scaling panel that connects to hardware for displaying and recording weight data; (c) a multimodal timer panel for timekeeping; (d) a dialogue panel for displaying and recording interactions with the AI assistant "Yummy"; and (e) an evaluation panel for assessing and recording the actual cooking results. Additionally, the interface comprises a step window for displaying cooking instructions, a recording panel for editing and reviewing cooking operations and a calorie panel for calculating the caloric of finished dishes.

Recipe panel. In the recipe panel (Figure 3-a), we combined theoretical knowledge from the literature [37] with controlled variable experimental methodologies to conduct practical experiments on the effects of different component ratios between sugar and butter to establish rules to realize the customization features. Users may achieve taste by directly adjusting component proportions numerically or manipulating the three flavour characteristics of "sweetness", "texture", and "milky aroma" to construct a DIY recipe. Also, users can examine real-time recipe previews generated by the Rasa model in the left window.

Scaling panel. Users can refer to the DIY recipe data on the recipe panel and measure or adjust the actual ingredient quantities based on the values displayed in the scaling panel (Figure 3-b). A triangular progress bar visualizes the weighing progress. Users can use voice commands to instruct the AI assistant "Yummy" to switch ingredients, record the actual measured amounts, or adjust

ingredient quantities. Alternatively, these operations can also be performed via the GUI.

Timer panel. The user can quickly start, pause, restart, or terminate the timer via voice command or GUI (Figure 3-c). When the countdown reaches zero, the AI assistant will send a verbal notification signalling the end of the timer.

Dialogue panel. BakeEase system presents the user's voice interaction with the AI assistant "Yummy" as a chat discussion on the dialogue panel to display and record real-time text, images, and videos during the cooking process (Figure 3-d). When the wake-up keyword "Yummy" is detected, the active AI assistant will respond with a vocal acknowledgement of "I'm listening" and the text "listening" on the dialogue panel. Users can also engage the AI assistant by manually tapping the speaking button.

Evaluation panel. On the evaluation panel (Figure 3-e), the data in "actual amount" section reflects the precise weight of each ingredient as measured by the kitchen scale during the user's cooking process, providing a reference for future repetitions or recipe updates. In the "review" section, users move the slider to rate the finished result. Simultaneously, capture and upload a photograph of the finished product in the "baking results" section on the left.

4 PRELIMINARY EVALUATION

4.1 Participants and Procedure

We recruited five baking enthusiasts from the public (4 females and 1 male, aged 22–34) to participate in BakeEase's user testing. The average self-assessed baking skill proficiency was 2.8 (1 = extremely unfamiliar, 3 = moderately proficient, 5 = highly proficient), and the average self-assessed baking knowledge was 2.2 (1 = novice, 3 = some knowledge, 5 = extensive knowledge). Participants signed an informed consent form and provided demographic information before receiving a brief introduction to BakeEase. They then baked cookies while researchers observed their behaviours and ensured safety. After the experiment, participants completed the SUS Questionnaire [1] and a 15-minute semi-structured interview. Two researchers analyzed the data using thematic analysis [5].

4.2 General Impression

Participants who used BakeEase found the system's user interface clear and well-designed (P10), recipe steps straightforward and easy to understand (P12), and the instructional flow well-structured (P11). The SUS usability score of the BakeEase system was 72.75 (SD = 4.67).

Customization. Most participants (N=4) adjusted ingredient quantities and proportions based on their taste preferences, while more experienced user P10 adjusted the recipe numerically using standard values. Participants found the customization feature to be very helpful: "... You can adjust them based on your preferences. But if you use recipes from the internet, you can only follow. If you don't fully understand the purpose of each ingredient, you can't DIY..." (P8).

Making. During baking, participants used voice interactions to navigate between steps, control the kitchen scale for weighing ingredients and recording measurements, and interact with the AI assistant. They appreciated the convenience voice interaction provided when their hands were occupied: "When pouring flour or

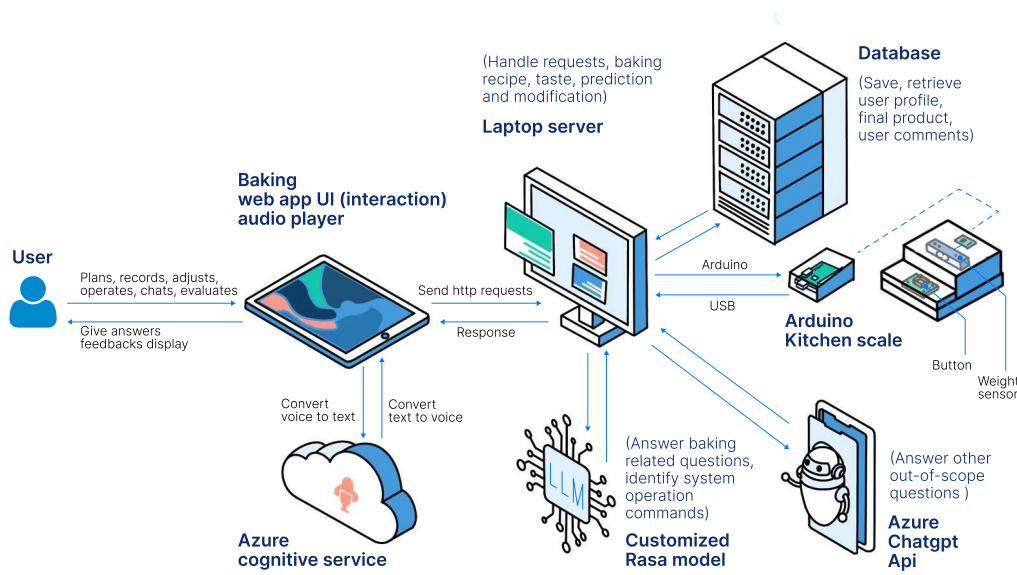


Figure 2: System Architecture

egg white... my hands were covered with egg white or flour... At times like these, the voice system was handy" (P11). Participants also expressed satisfaction with integrating the timer: "What I found particularly useful is... the timer" (P12). Feedback on using the AI assistant and kitchen scale is further detailed in Sections 4.3 and 4.4.

Evaluation. Participants found the Evaluation feature, which links actual baking data with taste evaluations, to be meaningful: "I think this is practical because every time you bake, there might be slight deviations, and the final taste could be different. This way, you can record it in time" (P8). P10 suggested further integrating note-taking features: "I'd like to have a notebook... where I can label each recipe as sweet or savory, and its type—whether it's cookies, cakes..." (P10). Participants (N=2) also noted that the calorie-tracking module would be helpful for individuals focused on fitness or weight management.

4.3 Interacting with the AI Assistant

During interactions with BakeEase's AI assistant "Yummy", participants found Yummy's suggestions to be helpful: "The comparison images or video demonstrations were generally convenient and engaging" (P12). However, not all participant questions fell within the dataset's scope (N=4). Integrating chatbot and LLM technologies allowed Yummy to address user-specific issues more effectively: "The problems you encounter are sometimes unique... when you ask Yummy immediately, it provides feedback promptly, which is great" (P8). When Yummy's responses were inconsistent with the actual context, participants noticed errors in the assistant's replies: "It mentioned baking powder just now (an ingredient not listed in the recipe), which was a bit confusing" (P10). Additionally, despite incorporating guidance on using tools in the dataset, Yummy still demonstrated

limitations in understanding the context. Baking often involves multiple tools, and participants needed more detailed guidance on artifacts to use at each step and how to use them effectively.

Another noteworthy issue was that participants were uncertain about how to interact with Yummy during their initial interactions, despite its brief self-introduction, which included helping with interface operations, weighing and recording ingredient amounts, and answering baking-related questions. "I couldn't figure out if it could understand my question" (P9). This uncertainty about Yummy's scope of capabilities and the level of assistance it could provide persisted until researchers explained its problem-solving features in detail: "I didn't know it could help me solve this issue, or that it could be so precise" (P12). Participants expressed a desire for Yummy to incorporate more proactive features and interact with users more actively. For example, P11 suggested Yummy could offer encouragement during repetitive tasks like mixing. Furthermore, participants hoped Yummy would be more intelligent, like recording the baking process (P11) and employing AI to track ingredient states and activity progress in real time. "For instance, I added the flour in the wrong step earlier... If it had a camera, it could see that and remind me of my mistake—it would be better" (P12). When questioned if using a camera would raise privacy concerns, P12 stated that she was not worried about privacy issues in the context of baking.

4.4 Experience with the Kitchen Scale

Participants appreciated the kitchen scale's ability to record the actual measured amounts of ingredients: "What surprised me the most was the adjustment of the actual amount recorded. I think this is a significant highlight because, when I was adding flour and egg whites earlier, I made some mistakes" (P11). During the weighing process, participants focused on the numerical value changes and

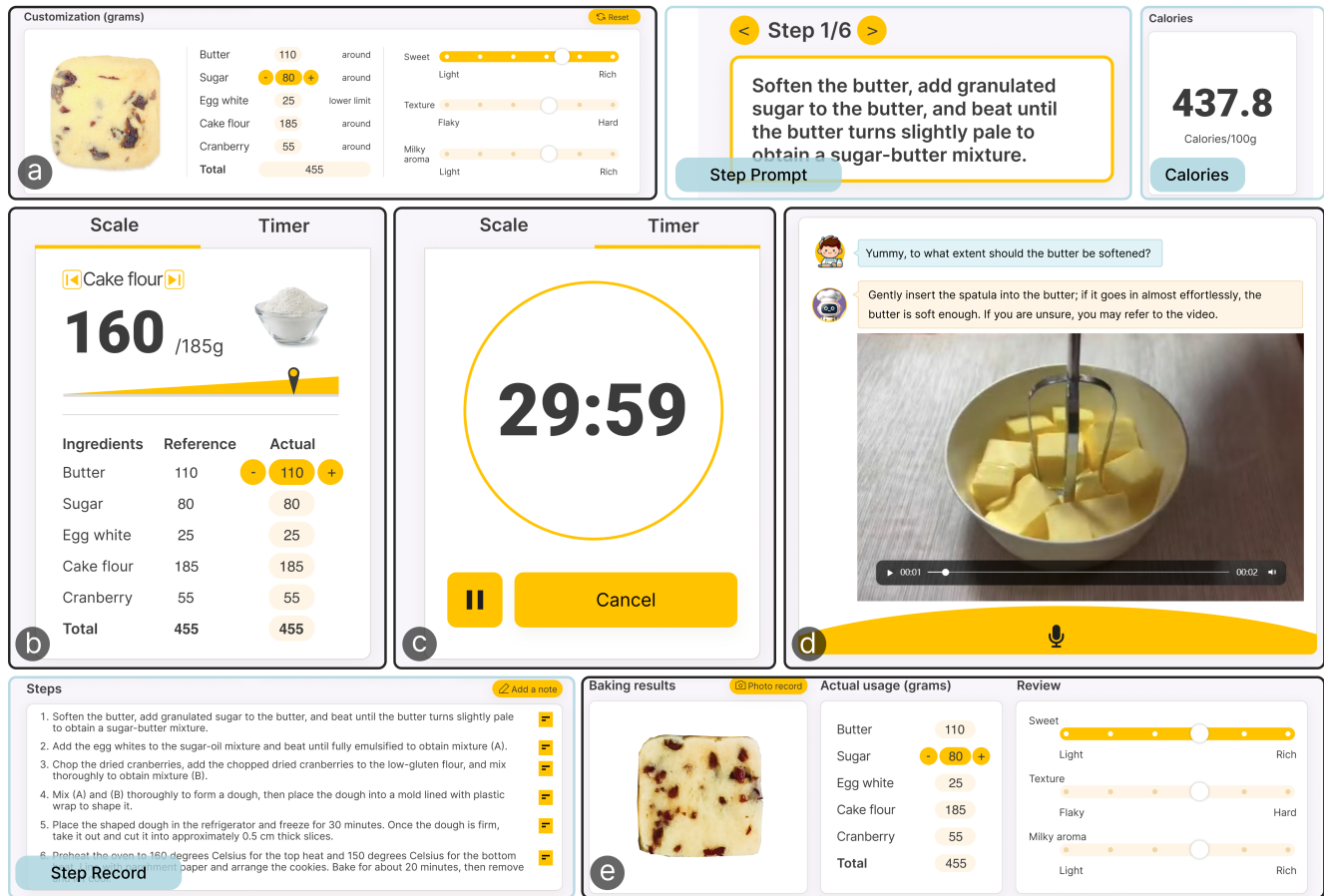


Figure 3: BakeEase Graphical User Interface Panels

did not notice the triangular progress bar, which was intended to indicate proximity to the target amount (P11). In such cases, audio prompts about weighing progress might more effectively balance user attention than relying on graphical indicators.

Participants also wanted the scale to have an automatic recording function (P9). Voice commands, while functional, added a degree of cognitive load: *"If I forget, it could be an issue... I'd have to remember it, which creates a bit of mental burden"* (P10). Because the weight values might continuously adjust or fluctuate during the weighing process, and without visual recognition, participants still needed to use voice commands to confirm the weighed ingredients. A potential solution is to add a physical button that allows users to quickly record the weight of the corresponding ingredient on the interface. This solution was deemed feasible after discussions with the participant (P11). However, P12 preferred voice interactions over physical buttons.

5 DISCUSSION

The findings from the preliminary evaluation indicate that our system demonstrates good adaptability to baking enthusiasts with varying skill levels and can meet their personalized needs. Users

with varying baking expertise and skill levels use the system at different frequencies. More experienced bakers interacted with the AI assistant less frequently, relying less on knowledge or skill-based advice, while the novice relied primarily on the assistant. For these users, video training before hands-on work was especially beneficial in building basic understanding. However, all users repeatedly stated a desire for voice-controlled interface interactions. Expanding the AI assistant's role to cover all phases of the baking process could further enhance its ability to assist users comprehensively.

In terms of automation, user preferences varied greatly. Some users preferred manual procedures, while others favoured automating repetitious jobs with machine aid. Regarding device connectivity, enabling interoperability—such as integrating mixers with automatic speed adjustments or controlling oven preheating—could help to streamline operations and save users time. Notably, the design goal of the BakeEase system should remain focused on supporting users in baking and enhancing their enjoyment of the activity rather than drawing users' attention to the system. To better facilitate user focus and reduce cognitive load, the interface layout could dynamically adjust based on user actions, highlighting the current task.

The design of the BakeEase integrates software and hardware, similar to Mimicook [30], enhancing the user experience after incorporating hardware. However, BakeEase holds an advantage by integrating an AI Assistant, which helps users expand their culinary knowledge. Compared to Google's Voice Assistant, which lacks context awareness and often generates responses unrelated to the task [18], BakeEase has specifically restricted the dialogue scope of the chatbot and LLM, ensuring that users gain task-relevant knowledge and improve their baking skills. Additionally, previous studies that focused on single-dimensional approaches [12, 21, 28]—which are primarily based on quantitative recipes and fixed workflows—BakeEase increases user autonomy, granting users greater creative flexibility in the baking process.

BakeEase system lays the groundwork for future development of context-aware cooking systems. It offers valuable insights for designing intelligent cooking systems. It has the potential for broader applications in general task guidance systems (e.g., crafts or DIY projects), skill development, and education (e.g., learning musical instruments, painting, or other step-by-step practices).

Limitations. Our research involved user studies with 7 participants and user testing with 5 participants, all based in China. We acknowledge the limitations posed by the small sample size and cultural specificity. Additionally, the living lab environment may not adequately represent the various aspects seen in real-world usage scenarios. Future studies should investigate real-world testing and include more diverse situations in experiments.

6 CONCLUSION AND FUTURE WORK

In this study, we explored user needs during the baking process and designed an AI-driven baking system prototype, BakeEase, which was preliminarily evaluated. The BakeEase system helps users complete baking tasks while enhancing their overall experiences. In the future, we plan to optimize the system's functionality and expand its versatility to accommodate a broader range of baking categories and ingredients. We will focus on leveraging user-generated data and images collected in the evaluation panels to enhance machine learning algorithms, enabling more personalized and adaptable recipe customization. It will enable baking enthusiasts to explore more creative possibilities of hands-on baking activity.

Acknowledgments

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